

Aparna Mane

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EDUCATION

K. J. Somaiya College of Engineering (Autonomous) — B.Tech, CGPA: 8.23/10 · Mumbai | India

June 2017 – June 2020

TECHNICAL SKILLS

Languages: C/C++, Python, Ruby on Rails,
APIs & Services: FastAPI, gRPC, REST API
Infra, Cloud & Data: Azure Cloud, DuckDB, Databricks, RabbitMQ, Redis, Kafka, Zookeeper
Databases: Cassandra, MySQL, MongoDB, PostgreSQL
AI/LLM: MCP servers, LLM agentic workflows, OpenAI API

PROFESSIONAL EXPERIENCE

Software Engineer · Hedgineer | Bangalore

Jan 2026 – Present

- **Designed** an autonomous Agentic Dispute Resolution engine using **Anthropic MCP** and FastAPI, enabling **LLM agents** to run real-time **DuckDB** ledger queries and autonomously triage chargebacks in real time.
- **Led** the re-architecture of the analytics layer from file-based Parquet storage to a **DuckDB OLAP** engine with in-memory CTEs, C++ vectorized execution, and secondary indexes, eliminating OOM risk on datasets of ~900 GB.
- Worked extensively with **DuckDB, Anthropic MCP, LLM agentic workflows, FastAPI, and Databricks** to build low-latency financial analytics and dispute-automation systems.

Associate → Manager → Director · Morgan Stanley | Mumbai

April 2022 – Jan 2026

- **Led** the design and delivery of a Kafka-based transaction processing pipeline handling **8 million events/day**, applying real-time transformations and persisting to **Delta Lake**, powering reliable downstream settlement reconciliation and analytics.
- **Owned** end-to-end development of a trade/cash settlement reconciliation engine, automating discrepancy detection across the settlement cycle and cutting manual investigation effort for the operations team.
- **Led** the migration from legacy Kerberos to **OAuth 2.0** authentication, strengthening compliance posture and delivering audit-ready authorization flows across platform APIs, in collaboration with the Security and Platform teams.
- Designed and implemented a Redis write-around cache for entitlement/authorization checks, reducing check latency from **203ms to 3.5ms** and improving responsiveness for downstream trading services.
- **Built** a configurable AI assistant (FastAPI + LLMs) integrated with real-time DAGs, collaborating with Trading and Risk teams to power live decision dashboards used by Strats & Traders daily.
- Worked extensively with **Kafka, Zookeeper, Delta Lake, Redis, OAuth 2.0, FastAPI, and distributed transaction systems** in a high-throughput financial trading environment.

Software Engineer · IDfy | Mumbai

Jan 2022 – April 2022

- **Redesigned** HRMS integration APIs and resolved a critical vendor hierarchy flaw in a background-verification flow, enforcing explicit parent-child verification at every level.

Associate Software Engineer · LogisticsNow | Mumbai

July 2020 – Jan 2022

- **Designed, deployed, and restructured** 200+ REST APIs and 80+ applications on **Azure Cloud**, improving performance and reliability for logistics transaction services.
- **Implemented** data **partitioning** and **MongoDB** replica sets to distribute transaction workloads and improve database availability under peak load.
- **Built** a real-time procurement auction/bidding portal using **SignalR** and **Redis**, streamlining supplier selection and reducing negotiation costs by up to **30%** for Fortune 500 clients.
- Worked extensively with **Azure Cloud, MongoDB, SignalR, Redis, and distributed transaction systems** to build and scale logistics platform services.

PROJECT HIGHLIGHTS

gRPC Microservices Integration | gRPC · Apache Flight · FastAPI · Protobuf · Python

- Replaced **REST** API with **gRPC** across internal services, cutting API latency from **3s to 600ms** and boosting throughput by **60%**, and integrated **Apache Arrow Flight** for zero-copy columnar transfer of large payloads.

On-Prem to Cloud Migration | FastAPI · Azure Kubernetes Service · OAuth2 · API Gateway

- Migrated **JupyterLab** and in-house **DzPy** data library from on-premises to **AKS**, replacing manual memory management with **auto-scaling** — reducing infrastructure costs by **30%** (\$11M to \$7.7M).

Auction Portal | Azure Functions · Cosmos DB · SignalR · Redis · Azure DevOps

- Built serverless REST APIs on **Azure Functions** with **Cosmos DB** for auto-scaling storage, integrated **SignalR** for real-time bidding updates and **Redis** caching for auction metadata, ensuring low-latency data retrieval.

CERTIFICATIONS & AWARDS

- **Claude Certified Architect – Foundations** | Introduction to Agent Skills | Building with Claude API Anthropic – May 2026
- **Microsoft Certified: Azure Fundamentals**, Microsoft – Feb 2024
- **IST COE Award** for Platform Modernization – Morgan Stanley, 2023
- **ESTAR Recognition Award** for Exceptional Delivery – Morgan Stanley, 2022